Research Statement

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Interpretability results increasingly function as safety evidence: if an internal representation can be located, steered, or ablated, we conclude that the behaviour it governs is understood — and controllable. My research asks what warrants that conclusion. The question grew out of philosophy of language: my ESSLLI 2025 paper argued that context-dependent semantics offers a substantive explanatory frame for transformer behaviour, and the invited follow-up (*under review*, *Journal of Logic, Language and Information*) tests that frame causally rather than descriptively. The through-line of my work since is to replace plausibility assumptions with falsifiable preservation tests.

Three results define the programme so far. **First,** sham-controlled activation patching shows causal effects of contextual token states on controlled disambiguation, counterfactual and coherence margins. Relevance-controlled effects vary across tasks and checkpoints, including a null contrast in the counterfactual test for Qwen2.5-1.5B. These experiments do not decide between semantic theories. **Second, a limitation result:** geometric fidelity does not certify causal preservation. In one Gemma-3-4B-PT task, error-preserving SAE delta patching recovered an estimated 0.621 of the mean raw donor-state replacement effect at L4 (95% CI 0.263–0.954). State-reconstruction FVU was 0.137 at L4. The estimated loss depended on the inclusion policy: the stricter diagnostic-gate interval included zero. At L8, the estimate was 0.974 (95% CI 0.749–1.320); equivalence was not established. **Third, a safety instantiation:** in the declared Gemma-3-4B-IT candidate search, none of 135 eligible directions passed the pipeline's applied KL≤0.1 selectivity gate: a bounded-search qualification failure. On 152 held-out HarmBench prompts, two Gemma-2-2B-IT layers with mean cosine similarity differing by 0.00134 recovered 0.335 versus 0.699 of a fixed ablation effect on a first-token refusal-score proxy (clipped log-odds; paired bootstrap 95% CI for the difference: 0.340–0.391). The layers were closely matched in cosine, not in centered FVU (9.633 versus 1.245); the comparison therefore concerns cosine as a preservation criterion. Scope limits include pretrained SAEs on instruction-tuned models in the Gemma-2 case and single-model cohorts.

As a PhD student I want to develop this into a **causal-preservation benchmark for white-box interpretability and control methods:** multiple model families and behavioural endpoints; multiple representation bases (SAE reconstructions and features, matched-rank PCA, random orthogonal projections, circuit- and attribution-based representations); pre-registered inclusion gates and matched interventions throughout; and robustness tested under context variation, ambiguity, and distribution shift. These results motivate testing which additional measurements and controls are needed for monitoring applications. Whether adversarial behaviour could evade those measurements remains a separate research question.

I run this programme independently end to end: reproducible intervention pipelines (TransformerLens, activation patching, bootstrap inference, deterministic runners with provenance manifests), built and executed on my own infrastructure. My formal background in logic and philosophy of language is not ornamental here — it is what keeps the evidence standards honest.

Gemma-3 provenance note: The local provenance does not fully establish that the human threshold decision preceded all candidate-table inspection.

SAE behavioral-preservation manuscript: Earlier draft on GitHub.